

Machine Learning-Based Pulse Shape Discrimination in Plastic Scintillators using Geant4 Simulations

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Abstract

Pulse Shape Discrimination (PSD) plays a central role in radiation detection by enabling reliable neutron–gamma separation. Plastic scintillation detectors are widely used in monitoring and security applications due to their robustness and scalability; however, their PSD performance is intrinsically limited by pulse distortions and noise arising from scintillation dynamics, photon-counting statistics, and optical photon transport effects. In this thesis, neutron–gamma discrimination in EJ-276 plastic scintillators is investigated using a fully simulation-based framework. Scintillation pulse formation is modeled using Geant4-based Monte Carlo simulations that incorporate optical photon generation, transport, and detection, resulting in time-resolved scintillation waveforms. The simulated signals are analyzed in the time domain with particular emphasis on noise sources originating from photon statistics and transport-induced temporal dispersion. Idealized reference pulse shapes are injected and figure in the analysis as pure shape factors to extract the intrinsic scintillation response. In this case, particle identification is based on classical PSD techniques (charge comparison method and FoM analysis) and a multilayer perceptron (MLP) trained directly with a digitized pulse waveform. The performance of classical methods is observed to degrade across the investigated neutron energy range of 1–4 MeV, where statistical fluctuations increasingly limit reliable neutron–gamma separation. In contrast, the MLP-based approach, which exploits the full time-domain pulse shape, achieves robust discrimination and yields an overall classification accuracy of approximately 93%. These results demonstrate that time-domain, shape-based analysis combined with machine learning provides a powerful and scalable alternative to conventional PSD techniques for neutron–gamma discrimination in plastic scintillation detectors.

1 Introduction

Pulse Shape Discrimination is a fundamental signal processing technique in radiation detection that enables the identification of different types of incident particles within mixed radiation fields. Its importance is particularly evident in applications including radiation monitoring systems and detector-based measurement environments, where weak neutron signals must be reliably distinguished from gamma-ray backgrounds [10]. By using differences in the time structure of detector signals, PSD improves particle identification and increases the reliability of radiation measurements under realistic conditions.

The physical basis of PSD lies in the interaction-dependent scintillation response of detector materials. In organic plastic scintillators, neutron–gamma discrimination arises from differences in the decay characteristics of scintillation light, even though the initial rising edges of the pulses are often similar. Gamma rays mainly interact through electromagnetic processes, exciting singlet molecular states that decay quickly via prompt fluorescence. Neutrons, in contrast, scatter elastically from hydrogen nuclei and produce recoil protons with dense ionization tracks. These interactions favor triplet states, resulting in delayed light emission that extends the pulse tail. As a result, neutron-induced pulses exhibit a systematically longer decay than gamma-induced pulses, providing the physical contrast exploited by PSD methods.

Despite this clear physical distinction, neutron–gamma separation in plastic scintillators remains challenging. Classical (statistical) discrimination techniques typically rely on scalar quantities derived from prompt–tail comparisons, such as charge comparison or tail-to-total ratios. While these methods are computationally efficient and physically interpretable, they depend strongly on a limited portion of the pulse waveform, most notably the delayed tail. Consequently, their performance degrades in the presence of pulse distortions, photon-statistical fluctuations, and transport-induced timing variations, all of which become more pronounced in large-volume detectors and realistic geometries.

Advances in Monte Carlo simulation have enabled increasingly detailed modeling of scintillation pulse formation by explicitly incorporating scintillation decay components, optical photon generation, and photon transport within complex detector geometries [4]. These studies have shown that the observed pulse shape is not a direct representation of intrinsic scintillation kinetics alone, but rather a convolution of emission processes and optical photon transport–induced temporal dispersion effects such as path-length differences, reflections, and boundary interactions. In this context, noise and pulse-to-pulse variability should be regarded as inherent features of the detection process that directly influence discrimination performance, particularly in the late-time regions where neutron–gamma separation is most pronounced. In this framework, the timing of

scintillation photon arrivals is a key element, as it reflects both the intrinsic scintillation decay and the effects of optical photon transport. Since neutron–gamma discrimination in plastic scintillators mainly relies on small differences in the late part of the pulse, accurately modeling photon arrival times is essential for realistic PSD studies.

Motivated by these considerations, this thesis adopts a time-domain approach to the analysis of scintillation pulse shapes. A fully simulation-based framework is developed using GEANT4 to generate event-by-event, time-resolved scintillation pulses in EJ-276 plastic scintillators. To separate intrinsic scintillation behavior from random fluctuations, idealized reference (gold) pulse shapes are used as stable representations of the underlying scintillation response. These reference pulses provide a baseline for shape-based analysis and for assessing how noise affects discrimination performance. Building on this representation, particle identification is carried out using a multilayer perceptron (MLP) trained directly on digitized pulse waveforms. The MLP is used within a physically motivated framework that maintains the temporal structure of scintillation signals, as opposed to treating the model as a black box. This method overcomes the drawbacks of traditional PSD techniques and increases robustness against noise and pulse distortions by utilizing the entire waveform rather than just one scalar parameter.

The study begins with a review of the physical processes underlying scintillation and pulse shape formation in organic plastic detectors. This is followed by a description of the GEANT4 simulation framework used to generate time-resolved scintillation pulses. The time-domain properties of the simulated signals are then examined, with particular emphasis on noise effects and the construction of reference pulse shapes. Finally, waveform-level particle identification using the MLP is developed and its performance is evaluated in comparison with classical pulse shape discrimination techniques.

2 Physical Background of Scintillation and PSD Techniques

It is particularly crucial in scenarios like neutron detection in gamma-dominated environments where a weak signal of interest needs to be found in the presence of a strong background. By separating signals based on their pulse shapes, PSD improves the reliability of radiation measurements and reduces background-related ambiguities. The basic idea behind PSD is that different particles produce detector signals with distinct time profiles. The decay region shows variations in how energy is deposited in the scintillator, whereas the rising part of the signal is frequently similar for various interactions.

The basic idea behind PSD is that different particles produce detector signals with distinct time profiles. While the rising part of the signal is often similar for different

interactions, the decay region reflects differences in how energy is deposited in the scintillator. These differences in the pulse tail carry information about the particle type and form the physical basis of pulse shape discrimination. Careful analysis of the pulse shape therefore allows the identification of the particle responsible for a given event, forming the basis of discrimination techniques.

This physical principle is most clearly demonstrated in neutron–gamma separation using organic scintillators. Gamma rays primarily interact with these materials through Compton scattering, producing excited electrons that excite singlet states in molecules. Prompt fluorescence causes these states to decay quickly, producing pulses with fast decay components. Conversely, neutrons interact primarily with hydrogen nuclei through elastic scattering, resulting in recoil protons that deposit energy along dense ionization tracks. The population of molecular triplet states, which relax via slower processes like delayed fluorescence and phosphorescence, is favored by this high ionization density. In contrast to gamma-induced pulses, neutron-induced signals have a significant slow component, which results in a longer pulse tail. The physical basis for neutron–gamma pulse shape discrimination in organic scintillators is this difference in decay characteristics.

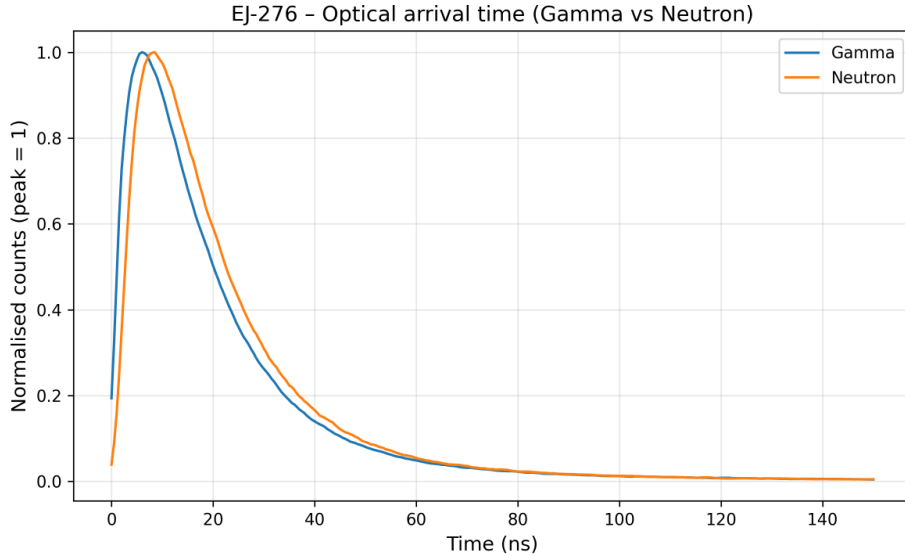


Figure 1: Example gamma-neutron pulse shape

Over time, numerous methods have been developed to quantify and exploit these pulse shape differences. Broadly, PSD techniques can be categorized into two conceptual frameworks: statistical discrimination methods and prior-knowledge-based discrimination methods. In statistical approaches, a discrimination parameter is calculated for each recorded pulse, and the resulting distribution of this parameter across many events is analyzed. These distributions often form distinct clusters, frequently approximated by Gaussian shapes, corresponding to different particle species. Classical time-domain

techniques such as Charge Comparison (CC) and Charge Integration (CI) fall within this category and have been extensively employed due to their conceptual simplicity and robustness. The purpose of this section is to summarize only the PSD concepts that are directly used in the analysis pipeline developed in this work.

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2.1 Classical Discrimination Methods

Among the most commonly used pulse shape discrimination techniques are charge-based methods, which exploit differences in the delayed scintillation components of neutron and gamma-induced signals. In organic scintillators, neutron interactions typically enhance the slow scintillation component, leading to an increased contribution of late-time light compared to gamma interactions. Classical charge-based PSD methods aim to quantify this difference through time-domain integration of the detector signal.

Charge comparison (tail-to-total) parameter. In this thesis, the charge comparison (CC) method is employed as a baseline classical pulse shape discrimination metric. The CC method quantifies neutron–gamma differences by comparing the amount of scintillation charge collected in prompt and extended time windows of a digitized pulse waveform.

For each pulse, the charge integrated within a short time window, Q_{short} , is compared to the charge accumulated within a longer time window, Q_{long} . The CC discrimination parameter is defined as

$$\text{CC} = \frac{Q_{\text{long}} - Q_{\text{short}}}{Q_{\text{long}}}. \quad (1)$$

Here, Q_{short} corresponds to the integrated charge within the first $T_{\text{short}} = 80$ ns, which is dominated by prompt scintillation light, while Q_{long} represents the integrated charge within $T_{\text{long}} = 350$ ns and includes both prompt and delayed scintillation components. Physically, the CC parameter measures the fractional contribution of late-time scintillation light relative to the total signal.

Neutron-induced pulses typically exhibit larger CC values due to their enhanced slow scintillation component, whereas gamma-induced pulses are dominated by prompt emission and therefore yield smaller CC values. Since the neutron–gamma pulse shape difference is distributed across the decay tail rather than confined to a narrow time interval, the CC method captures the cumulative delayed contribution of the scintillation signal. However, the method remains sensitive to statistical fluctuations in the tail region and to temporal broadening effects introduced by optical photon transport, which can degrade discrimination performance, particularly at low energies.

The selection of the integration time windows is guided by the temporal structure

of the scintillation pulse. The short integration window T_{short} is chosen such that it captures the majority of the prompt scintillation component, typically corresponding to approximately 90% of the fast light output. This ensures that Q_{short} is primarily sensitive to the prompt emission and minimizes contamination from delayed scintillation components.

The long integration window T_{long} is selected in a range where the cumulative pulse integral reaches saturation, providing a stable measure of the total deposited energy. In plastic scintillators, this saturation typically occurs within a time interval of 300–500 ns. Consequently, T_{long} is chosen to fully include both prompt and delayed scintillation light while avoiding unnecessary integration of baseline noise.

Identification of the delayed scintillation region. The onset of the delayed scintillation component is identified by examining the time-dependent difference between the average neutron and gamma pulse shapes. This comparison provides guidance for selecting a tail integration window that starts after the prompt component has largely decayed and covers the region where neutron–gamma differences persist.

Figure of Merit (FoM). For a scalar PSD parameter, the separation performance between neutron- and gamma-induced events is commonly summarized using the Figure of Merit (FoM). The FoM quantifies the degree of overlap between the two populations in the discrimination-parameter space and is defined as

$$\text{FoM} = \frac{|\mu_n - \mu_\gamma|}{\text{FWHM}_n + \text{FWHM}_\gamma}, \quad (2)$$

where μ_γ and μ_n denote the centroid of the gamma and neutron PSD-parameter distributions, respectively, and FWHM_γ and FWHM_n are their corresponding full widths at half maximum (FWHM). Larger FoM values indicate clearer separation and reduced class overlap, whereas smaller FoM values reflect increased overlap between the two particle classes.

In this thesis, μ_γ and μ_n are extracted from the PSD histograms, and the widths are characterized by the FWHM of each distribution, ensuring a compact, physically interpretable scalar summary of separation quality. Charge-based PSD techniques are computationally efficient and physically interpretable, but they compress each scintillation pulse into a single scalar quantity derived from predefined integration gates. As a consequence, discrimination performance can degrade under realistic conditions where the late-time tail region is strongly affected by photon-statistical fluctuations and transport-induced temporal dispersion. These limitations motivate waveform-level approaches that operate directly on time-binned pulse shapes, which are developed in later sections of this thesis.

3 Simulation and Data Generation

In order to study neutron–gamma pulse shape discrimination, a Monte Carlo simulation framework was developed using GEANT4. The simulation explicitly models scintillation photon generation and optical photon transport in an EJ-276 plastic scintillator, allowing the formation of time-resolved pulse waveforms that directly reflect both intrinsic scintillation dynamics and transport-induced timing effects.

3.1 Detector Setup with Geant4 Simulation Framework

GEANT4 is an Monte Carlo simulation toolkit for the simulation of the passage of particles through matter [1]. It provides a validated framework for simulating hadronic and electromagnetic processes, as well as optical photon generation and tracking, making it particularly suitable for nuclear physics and scintillation detector studies.

The EJ-276 plastic scintillator has been selected as the detector material as it can distinguish between different neutron–gamma pulse shapes while maintaining the scalability and mechanical durability of organic plastics. The manufacturer datasheet provided by Eljen Technology [2] was used to determine the scintillation properties of EJ-276, including particle-dependent decay components, light yield, and some optical parameters. In contrast to traditional plastic scintillators, EJ-276 includes improved slow scintillation components whose relative contribution varies depending on the kind of interacting particle. This results in measurable variances in the temporal structure of pulses caused by gamma and neutrons. EJ-276 shows better radiation hardness and long-term stability than other PSD-capable plastics like EJ-299-33, which makes it more suitable for this thesis [6].

To investigate neutron–gamma pulse differences in EJ-276 plastic scintillator, a dedicated GEANT4 simulation implemented following the methodology adopted by [6]. An improved time-constant model, in which fast, intermediate, and slow decay channels are populated with varying relative weights for neutron and gamma interactions, was used to explicitly add particle-dependent scintillation decay components. Neutron interactions primarily favor the slower decay channels, whereas gamma interactions mostly stimulate the fast component, resulting in distinct variations in the temporal structure of the emitted scintillation light.

The detector geometry consists of a cubic EJ-276 scintillator with fixed dimensions of $10 \times 10 \times 10 \text{ cm}^3$. To maintain isotropic symmetry, a volume geometry was used rather than a slab design. Equal amounts of gamma rays and neutrons with defined kinetic energy made up the primary particles, which were produced from the center of the scintillator with isotropic angular distributions.

All external faces of the scintillator were covered with thin ($10 \text{ }\mu\text{m}$), highly reflective Teflon layers, shown in red in the GEANT4 visualization, to enhance internal reflection

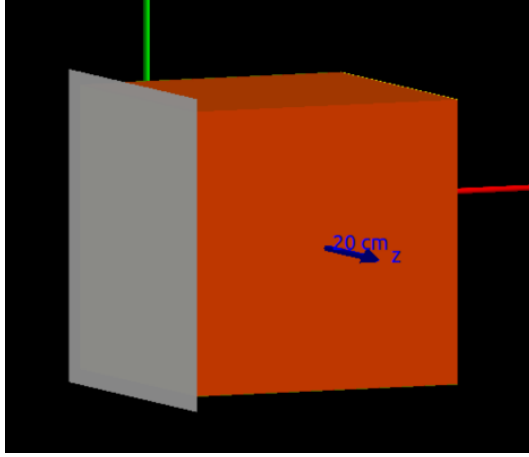


Figure 2: GEANT4 visualization of the cubic $10 \times 10 \times 10 \text{ cm}^3$ EJ-276 scintillator geometry prior to particle injection.

and prolong optical photon propagation within the volume. On the negative x -face of the scintillator, the Teflon coating was removed and replaced by an aluminum plate, shown in gray, which serves as the effective photodetector surface. To minimize optical photon losses observed in preliminary simulations, the aluminum plate was defined with slightly larger lateral dimensions than the scintillator face, thereby photon escape through small geometric gaps at the boundary has been reduced.

3.2 Optical Photon Tracking and Pulse Construction

Considering that the discriminatory information is stored in the relative timing of scintillation light emission rather than solely in its total intensity, pulse shape discrimination is inherently a time-domain problem. Intrinsic scintillation decay kinetics, together with optical photon transport-induced temporal dispersion effects (i.e. path-length variations, surface reflections, and boundary interactions) shape the electrical signal observed in a detector. These processes introduce event-to-event timing fluctuations that become increasingly significant in large-volume scintillators. For this reason, the present simulation framework explicitly records the arrival times of individual scintillation photons at the aluminum photodetector surface. Optical photon tracking was enabled for all events, and only photons reaching the detector surface were registered.

This strategy allows realistic time-domain pulses, containing both prompt and delayed components relevant for neutron–gamma discrimination, to be constructed directly from the simulated data. For each detected optical photon, the arrival time at the detector surface was stored. To construct time-domain scintillation pulses, the time axis was discretized into fixed bins, and photon arrival times were accumulated into event-by-event histograms. Thus, each event is represented by a time-binned waveform vector

$$\mathbf{x} = [n_1, n_2, \dots, n_N], \quad (3)$$

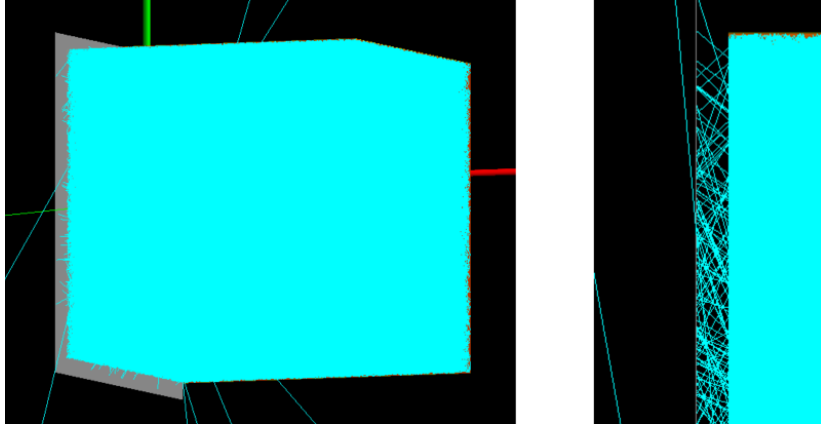


Figure 3: i) Distribution of optical photons generated in the scintillator following particle injection (left). ii) Close-up view of optical photons reaching the aluminum photodetector surface (right).

where n_k denotes the number of detected photons in the k -th time bin. In this work, the number of timebin was fixed to $N = 5000$ with a bin width of $\Delta t = 0.1$ ns, corresponding to a total acquisition window of 500 ns. This representation preserves both prompt and delayed scintillation components and forms the basis for all subsequent pulse shape analyses.

3.3 Synthetic Dataset Generation and Physical Validation

The time-binned scintillation pulses obtained through optical photon tracking form a fully synthetic dataset generated using GEANT4. Each simulated event corresponds to a single primary particle injected at the center of the scintillator and is represented by a waveform encoding the temporal evolution of detected scintillation light at the aluminum photodetector surface. For every event, the dataset stores the time-binned pulse waveform together with the corresponding particle label (neutron or gamma). By constructing the dataset at the waveform level, the full temporal structure of the signal is preserved, including both prompt and delayed scintillation components that are essential for neutron–gamma pulse shape discrimination.

The present simulation focuses exclusively on scintillation photon generation and optical photon transport. Effects related to the electronic response of the photodetector and readout chain, such as transit-time spread, digitization effects, were intentionally not included. This choice ensures that the generated pulses maintain a direct correspondence with the underlying scintillation and optical transport physics, providing a clean and physically transparent dataset for methodological studies.

Physical validation of the synthetic dataset was performed by comparing neutron- and gamma-induced pulse shapes. Consistent with previous experimental and simulation-based studies, neutron-induced pulses exhibit a more pronounced slow component, while gamma-induced pulses decay more rapidly. This difference appears as an enhanced

tail structure in neutron pulses and confirms that the simulated signals reproduce the essential physical features underlying classical pulse shape discrimination. For comparison with traditional PSD techniques, scalar quantities such as total pulse integral and tail-to-total ratios can be derived from the same waveforms. These quantities are used solely for validation and benchmarking purposes, while the classification models presented in later chapters operate directly on the full time-binned pulse shapes.

3.4 Dataset Design and Energy Scan Strategy

To evaluate the robustness and generalization capability of neutron–gamma pulse shape discrimination, the synthetic dataset was generated at multiple discrete particle energies within the MeV range. Training and evaluating discrimination models at a single fixed energy can lead to overly optimistic performance estimates, as the model may implicitly exploit energy-specific pulse characteristics. To avoid this bias, an energy-separated dataset design was adopted, in which training, validation, and testing are performed at different energies.

Neutron and gamma events were simulated at seven discrete kinetic energies: 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, and 4.0 MeV. At each energy point, a total of 4000 events were generated, consisting of 2000 gamma-ray events and 2000 neutron events. Rather than randomly mixing events from all energies, the dataset was partitioned by energy to enforce a physically meaningful separation between training and evaluation conditions. Events generated at 1.5, 2.0, 2.5, 3.0, and 3.5 MeV were used exclusively for training, providing representative coverage over the intermediate energy range. Validation was performed using a fully held-out low-energy dataset at 1.0 MeV, enabling model selection under an explicit energy shift relative to the training conditions. Final performance evaluation was carried out on an independent test dataset at 4.0 MeV, which lies outside the training energy range. With this energy-held-out design, model performance can be evaluated under controlled energy shifts without information leakage across subsets. The resulting dataset configuration is summarized in Table 1.

The selected energy range focuses on the MeV regime, where neutron–gamma pulse shape discrimination in organic scintillators is strongly influenced by photon statistics and optical transport effects, providing a sensitive testing ground for evaluating the robustness of discrimination methods. Although the simulation framework is capable of generating datasets at much lower (keV-scale) or higher (tens of MeV) energies, these regimes were not included in the primary analysis. At very low energies, extremely limited photon statistics would dominate pulse variability and obscure intrinsic shape differences. The chosen energy range therefore represents a balanced and physically relevant test case. Extensions to other energy regimes remain straightforward within the present framework and may be explored in future studies.

4 Time-Domain Pulse Shape Analysis and Noise Effects

Scintillation pulses produced by neutron and gamma interactions exhibit characteristic differences in their temporal structure that form the physical basis of pulse shape discrimination. While these differences originate from particle-dependent excitation and decay mechanisms, the observable pulse shape is further modified by photon-counting statistics and by time dispersion introduced during optical photon transport inside the detector volume. As a result, neutron–gamma separation is governed not only by intrinsic scintillation decay constants, but also by how reliably these decay features are preserved in the time-domain representation of the signal.

A time-domain analysis therefore provides direct insight into both the origin of discrimination features and the mechanisms that degrade them. By examining pulse morphology, identifying dominant sources of temporal variability, and introducing reference pulse shapes as a stable baseline, it becomes possible to clarify why classical charge-based discrimination parameters lose effectiveness under realistic conditions and to motivate waveform-level approaches that exploit the full temporal information in scintillation signals.

4.1 Pulse Shape Characteristics in the Time Domain

From a time-domain perspective, scintillation pulses can be characterized by a small number of physically meaningful descriptors, including the rise time, the overall decay profile, and the relative strength of the late-time tail. The rise time is primarily governed by the prompt scintillation component and photon collection efficiency, and is typically similar for neutron- and gamma-induced events in organic scintillators. In contrast, differences in decay behavior emerge predominantly at later times, where neutron interactions populate slow scintillation channels more strongly than gamma interactions.

At lower deposited energies, photon-counting noise and transport-induced timing dispersion lead to larger event-to-event variations in pulse shape. These fluctuations primarily affect the late-time tail region, which carries most of the discrimination-relevant information, and result in reduced separation performance. In contrast, at higher energies, increased photon statistics suppress relative fluctuations and produce more stable pulse shapes. Figure 4 shows the normalized photon arrival-time distributions for neutron- and gamma-induced events at

deposited energies of 1, 2, 3, and 4 MeV. While the overall pulse morphology remains similar across this energy range, neutron-induced signals consistently exhibit a more pronounced late-time tail compared to gamma-induced signals. This behavior reflects differences in energy deposition density and scintillation decay processes in organic

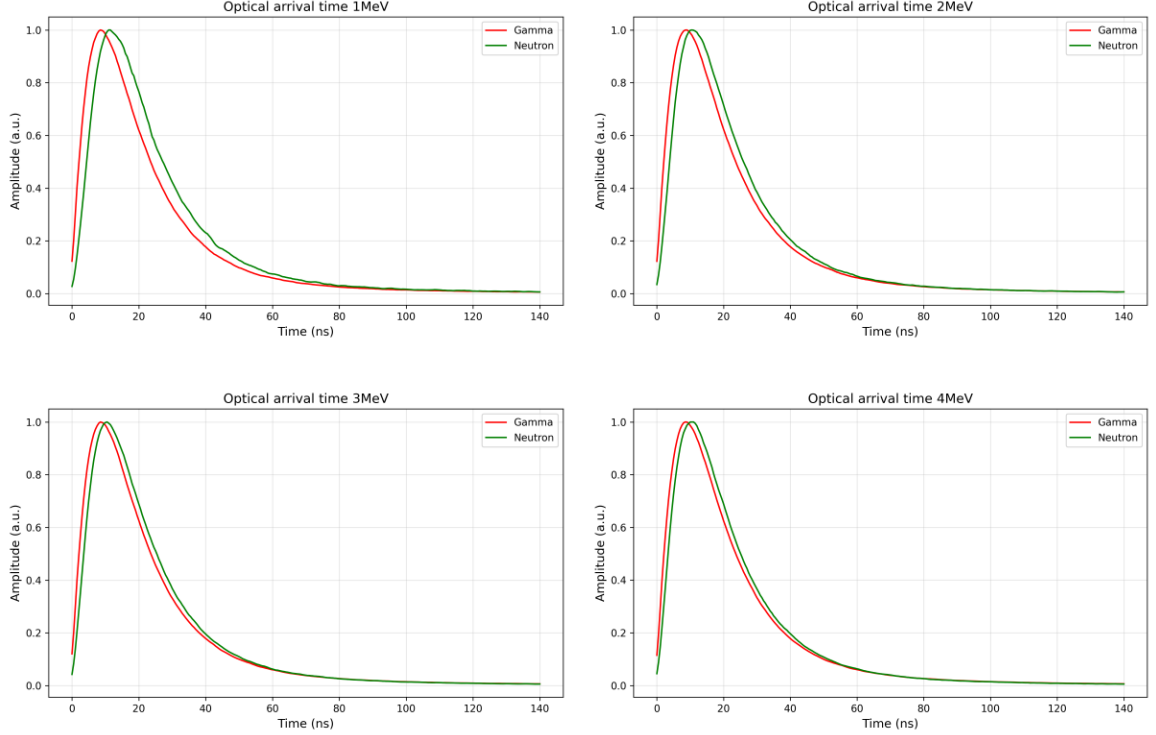


Figure 4: Normalized photon arrival-time distributions for neutron- and gamma-induced events at energies of 1, 2, 3, and 4MeV.

materials.

As the deposited energy increases, the contrast between neutron- and gamma-induced decay profiles gradually decreases. At higher energies, increased photon statistics suppress relative fluctuations in the tail region for both particle types, resulting in narrower pulse-shape distributions. At lower energies, however, reduced photon-counting statistics, transport-induced path-length variations, and reflections at detector boundaries introduce significant timing fluctuations. These effects broaden the distribution of tail features and increase the overlap between neutron- and gamma-induced pulses, making discrimination based solely on late-time pulse properties more challenging. While pulse shape differences provide useful physical insight into neutron–gamma separation, quantitative evaluation requires a scalar metric that captures the degree of class separation achieved by a given method. In classical pulse shape discrimination, this role is commonly fulfilled by the FoM, which is used here as a baseline metric to quantify the overlap between neutron and gamma charge comparison distributions through their relative peak positions and widths.

4.2 Classical Charge-Based PSD as a Diagnostic Tool

To quantify the impact of time-domain pulse broadening on neutron–gamma separation, a classical charge-based pulse shape discrimination parameter is employed as a diagnostic tool. The charge comparison (CC) method and the associated FoM are used in the

form defined in Section 2.1, serving here not as the primary classification technique but as a physically interpretable benchmark. For each event, the simulated waveform is integrated over predefined total and tail time gates, yielding a scalar CC parameter. The resulting CC distributions for neutron and gamma events provide a direct measure of how statistical fluctuations and transport-induced timing variations translate into class overlap within a traditional PSD framework. The separation performance is summarized using the FoM, as defined in Section 2.1. In this context, any reduction in FoM can be interpreted as a direct consequence of time-domain pulse distortions, rather than a failure of the underlying scintillation physics. The FoM therefore provides a convenient scalar proxy for quantifying how noise and optical transport effects degrade classical discrimination performance.

The integration gates were defined relative to the pulse maximum using fixed bin offsets consistent with the waveform discretization. The total gate spans from 10 ns before the peak to 300 ns after the peak, while the tail gate starts 20 ns after the peak and extends to the end of the acquisition window. These observations motivate the transition from scalar, gate-based discrimination to waveform-level approaches that exploit the full temporal structure of scintillation pulses, as developed in the following section. The charge-integration windows were defined relative to the pulse maximum. In the CC implementation used throughout this work, a short integration window of $T_{\text{short}} = 80$ ns captures the prompt component, while a long window of $T_{\text{long}} = 350$ ns integrates the pulse up to the delayed region. These values were selected by scanning candidate window pairs and choosing the setting that maximized separation on the available energy bins, and the same fixed windows were then used consistently for all CC evaluations reported in this thesis.

4.3 Reference (Ideal) Pulse Shapes and Noise Identification

To disentangle intrinsic pulse shape differences from event-level fluctuations, reference (*gold*) pulse shapes were constructed from the simulated dataset. As demonstrated above, the pulse tail carries the strongest discrimination-relevant information but is also the region most strongly affected by statistical fluctuations and transport-induced timing variations. Reference pulses therefore provide a stable baseline that isolates particle-dependent scintillation behavior in a noise-suppressed form.

In this work, reference (gold) pulse shapes are defined as the peak-aligned and area-normalized averages of a large number of neutron- or gamma-induced waveforms, providing a noise-suppressed representation of the intrinsic scintillation response for each particle type. To construct the reference pulses, neutron- and gamma-induced waveforms were first aligned in time by shifting each pulse such that its maximum occurred at a common reference bin. This alignment removes trivial timing offsets while preserving the relative temporal structure of the decay. Each waveform was then

normalized to unit area, ensuring that subsequent comparisons focus exclusively on pulse shape differences rather than absolute light yield or deposited energy. Following alignment and normalization, reference pulse shapes were obtained by averaging over a large number of events belonging to the same particle class. This averaging procedure suppresses uncorrelated fluctuations arising from photon-counting statistics and optical photon transport, while retaining stable, class-dependent features of the pulse shape. In particular, the resulting gold pulses exhibit a smooth and well-defined decay profile, with neutron-induced pulses showing a consistently enhanced and stable tail component compared to gamma-induced pulses.

A direct comparison between an individual event waveform and its corresponding gold pulse provides a quantitative measure of event-level deviations. Large deviations indicate strong contributions from noise and transport-induced temporal dispersion, while smaller deviations correspond to cleaner pulses that closely follow the intrinsic scintillation response. This effect is most pronounced in the late-time tail region, where neutron–gamma discrimination originates but where timing fluctuations also have the largest impact.

As a consequence, event-level broadening of the pulse tail leads to increased variability in tail-integrated quantities. When compressed into a single scalar parameter, such as the charge comparison (CC) metric, this variability manifests as a broadening of the CC distributions for both particle classes. The increased overlap between neutron and gamma distributions directly reduces the FoM, even though the underlying scintillation physics remains unchanged. These observations demonstrate that discrimination-relevant information is distributed across the full decay profile of the scintillation signal and that classical gate-based methods are inherently sensitive to noise-induced temporal distortions in the tail region. This motivates the transition from scalar, integration-based PSD parameters to waveform-level classification approaches that operate directly on time-binned pulse shapes, as developed in the following section.

5 Shape-Based Particle Identification

In this study, a multilayer perceptron (MLP) architecture is employed to perform shape-based neutron–gamma discrimination directly on raw time-domain pulse waveforms generated using GEANT4 simulations. Unlike experimental approaches relying on measured detector signals, the present work is based entirely on simulated datasets, allowing full control over particle type, incident energy, and detector response.

The input to the network consists of digitized scintillation pulses represented as fixed-length vectors of time bins. This design choice is motivated by the physical nature of pulse shape discrimination, in which discrimination-relevant information is distributed across the full decay profile rather than being localized within a narrow

tail window. By avoiding handcrafted feature extraction, the MLP is allowed to learn intrinsic correlations between pulse morphology and particle type in a transparent and physically interpretable manner. Specifically, each input waveform corresponds to the detected optical photon counts as a function of time, represented as a high-dimensional vector $\mathbf{x} = [n_1, n_2, \dots, n_{5000}]$, where each element denotes the number of detected photons within a fixed time bin.

For comparison with classical pulse shape discrimination, the charge comparison (CC) method introduced in Section 2.1 is applied to the same dataset. While CC relies on a physically motivated tail-to-total integral ratio, it compresses the entire waveform into a single scalar parameter. In contrast, the MLP operates directly on the complete time-domain pulse shape, exploiting information distributed over both prompt and delayed scintillation components. As a consequence, CC-based discrimination is inherently limited by the loss of information associated with this dimensionality reduction, particularly in the presence of statistical fluctuations affecting the pulse tail region.

5.1 Preprocessing and dataset strategy

Following baseline subtraction, negative excursions are removed and each waveform is peak-aligned to a common reference time t_0 , defined as the sample index corresponding to the maximum amplitude. Peak normalization is then applied so that the classification task emphasizes pulse *shape* rather than absolute amplitude, enabling the model to learn particle-dependent decay characteristics across different energies.

To ensure sufficient statistical coverage of gamma-induced pulse shape variability, a larger number of gamma events is simulated compared to neutron events at the same incident energies. This choice reflects the higher gamma background rates expected in realistic detector environments and improves the stability of waveform-level statistics. The resulting class imbalance is intentionally introduced at the dataset generation stage and is explicitly handled during model optimization using a reweighted loss function.

The training dataset contains pulses generated at energies $\{1.5, 2.0, 2.5, 3.0, 3.5\}$ MeV, while validation and testing are performed at 1.0 MeV and 4.0 MeV, respectively. This energy-separated strategy ensures that model performance is evaluated under conditions not directly seen during training, providing a stringent test of generalization.

5.2 MLP model and training setup

The multilayer perceptron (MLP) architecture employed in this work is designed to operate directly on raw time-domain scintillation pulses for waveform-level neutron–gamma discrimination.

Input layer. The input layer represents the digitized time-domain pulse samples and serves as the interface between the scintillation signal and the neural network.

Table 1: Energy configuration used for training, validation, and testing the MLP model. Each energy point contains 4000 events in total, consisting of 2000 gamma-ray and 2000 neutron events.

Energy (MeV)	Usage
1.0	Validation
1.5	Train
2.0	Train
2.5	Train
3.0	Train
3.5	Train
4.0	Test

Each input to the network consists of a digitized scintillation pulse represented by the detected optical photon counts as a function of time, $\mathbf{x} = [n_1, n_2, \dots, n_{5000}]$, where each element corresponds to a discrete time bin. For each incident energy, the dataset contains 2000 gamma-ray and 2000 neutron-induced pulses.

Hidden layers. Two fully connected hidden layers with 32 neurons each are employed, using ReLU activation functions to capture nonlinear variations in pulse shape. By processing the complete waveform, the MLP is able to exploit discrimination-relevant information distributed across the entire pulse, including fast and slow scintillation components, pulse tail slope, overall pulse width, and late-photon statistical fluctuations. Model optimization is performed using the Adam optimizer.

Output layer and training procedure. The output layer consists of a single neuron producing a logit, which is mapped to a class probability through a sigmoid activation function. The resulting output value is interpreted as the probability of the event being gamma-induced, while values below the decision threshold correspond to neutron-induced events. A standard decision threshold of $\tau = 0.5$ is used for hard classification. Model training is performed using a reweighted binary cross-entropy loss function. Class imbalance is handled by assigning a positive-class weight of $w = 3.17$ (gamma taken as the positive class), corresponding to the ratio of neutron to gamma events in the training dataset. Early stopping with a patience of 10 epochs is applied to prevent overfitting, and the final model is selected based on the validation ROC-AUC.

Receiver Operating Characteristic (ROC) curves are used to evaluate discrimination performance in a threshold-independent manner by plotting the true positive rate (TPR) against the false positive rate (FPR) as the decision threshold is varied. The area under the ROC curve (AUC) provides a compact and robust scalar summary of classification performance, particularly suited for imbalanced datasets.

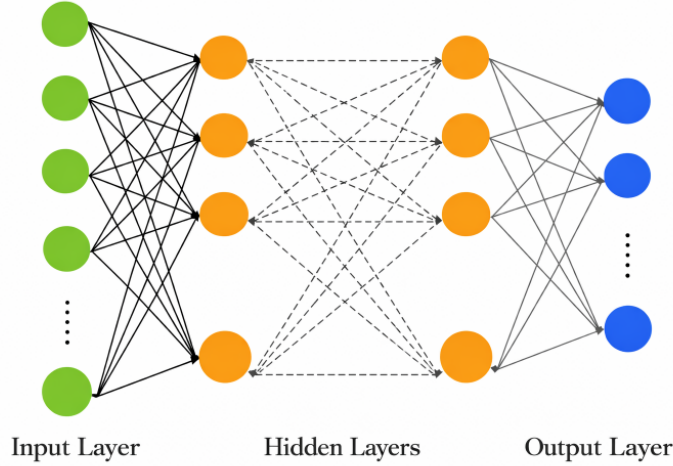


Figure 5: Schematic illustration of the multilayer perceptron (MLP) architecture used for waveform-level neutron–gamma discrimination.

6 Results and Validation

The trained MLP model is evaluated on an independent test dataset generated at 4.0 MeV, while training and validation are performed at $\{1.5, 2.0, 2.5, 3.0, 3.5\}$ MeV and 1.0 MeV, respectively. Using a fixed decision threshold of $\tau = 0.5$, the model achieves a test ROC-AUC of 0.980 and an overall accuracy of 93%. The confusion matrix indicates that neutron identification remains highly robust, while most misclassifications arise from gamma events being labeled as neutrons.

Table 2: Classification report of the test dataset at 4.0 MeV using the MLP model ($\tau = 0.5$).

Class	Precision	Recall	F1-score
gamma-ray	0.84	0.83	0.84
neutron	0.96	0.96	0.96

Figure 6 below shows the ROC curves obtained by the MLP classifier for different incident energies, while Fig. 7 presents a magnified view of the low-FPR region. Across the investigated energy range from 1.0 to 4.0 MeV, the classifier consistently achieves high discrimination performance, with ROC-AUC values ranging between 0.976 and 0.985. As expected, a gradual degradation in performance is observed at lower energies, where photon-statistical fluctuations and transport-induced temporal dispersion become more significant. Nevertheless, the waveform-level MLP maintains a strong separation between neutron and gamma events even in this regime. The zoomed ROC representation highlights the classifier performance in the experimentally relevant low-FPR region. The clear separation between curves in this region confirms the superiority of waveform-level

classification over classical gate-based methods, which typically suffer from significant performance loss under identical conditions.

To study the dependence of discrimination performance on deposited energy, ROC curves are additionally evaluated separately for each energy bin. As expected, discrimination degrades at lower energies due to increased relative impact of photon-statistical fluctuations and transport-induced temporal dispersion. For compact comparison, the AUC values extracted from the per-energy ROC curves are also plotted as a function of energy.

6.1 Energy Dependence and Comparison with Charge-Based PSD

From 1.0 to 4.0 MeV, charge-comparison (CC)-based discrimination exhibits a clear energy dependence, with ROC curves systematically shifting upward as the incident energy increases. This behavior reflects the reduced relative impact of statistical fluctuations at higher energies, where the delayed scintillation component is sampled with improved photon statistics. The deterioration of CC performance at low deposited energies originates from the fundamental reliance of the method on late-time pulse integrals. At low energies, the total number of detected scintillation photons is reduced, increasing the relative contribution of photon-counting statistics and transport-induced timing fluctuations. Since the CC parameter compresses the entire waveform into a single tail-to-total ratio, these fluctuations directly broaden the neutron and gamma distributions, leading to increased overlap and a reduced Figure of Merit.

In contrast to classical charge-based approaches, the MLP operates directly on the complete time-domain pulse shape rather than a predefined integration window. By processing the full waveform, the network exploits discrimination-relevant information distributed across both prompt and delayed scintillation components, including correlations that are inaccessible to single-observable methods. As a result, the MLP is less sensitive to localized distortions in the tail region and maintains stable discrimination performance even when individual time bins are affected by statistical fluctuations.

Figure 6 presents the ROC curves obtained with the MLP classifier for different incident energies. Across the investigated energy range from 1.0 to 4.0 MeV, the ROC-AUC values remain consistently high, indicating robust neutron–gamma separability. Figure 7 shows a magnified view of the low false positive rate (FPR) region, which is particularly relevant for neutron detection in high gamma-background environments. The zoomed ROC representation reveals a clear energy dependence of classifier behavior at small FPR values.

At lower energies, increased waveform variability arising from limited photon statistics and stochastic optical photon transport leads to a reduction in true positive rates at fixed false positive levels. As the deposited energy increases, improved photon statistics

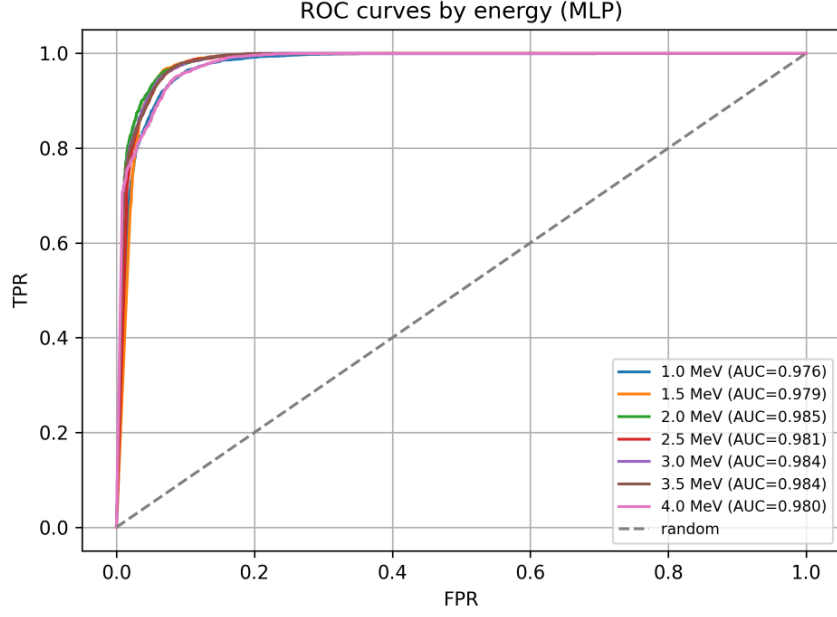


Figure 6: ROC curves obtained by the MLP classifier for different incident energies. The dashed line indicates the performance of a random classifier.

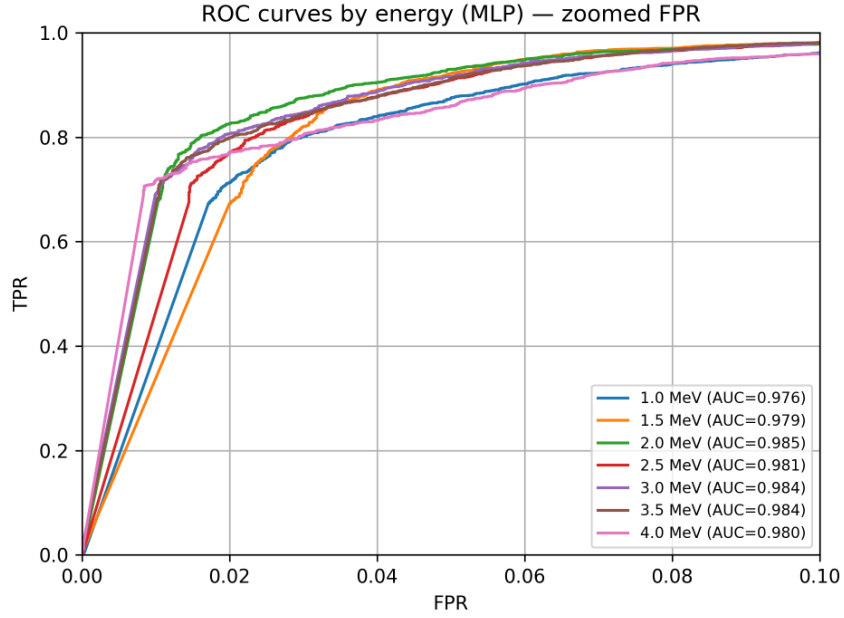


Figure 7: Zoomed view of the ROC curves shown in Fig. 6, focusing on the low false positive rate (FPR) region. The separation between energy-dependent curves is most pronounced at small FPR values, which is the operationally relevant regime for neutron detection in high gamma-background environments.

suppress relative timing fluctuations, resulting in converging ROC curves and more uniform classification performance across energies. A direct comparison with classical charge-based pulse shape discrimination highlights the contrasting energy dependence of the two approaches. FoM obtained from charge comparison decreases markedly at lower energies due to enhanced photon-statistical fluctuations and transport-induced broadening of the pulse tail. In contrast, the ROC-AUC of the waveform-level MLP remains relatively stable across the same energy range, indicating that the network exploits discrimination-relevant information distributed over the full temporal structure of the scintillation pulse rather than relying on a single tail-integral parameter. Overall, while CC-based discrimination suffers from a pronounced degradation at low energies, the MLP preserves high discrimination power under identical conditions. This result demonstrates that leveraging the full pulse shape provides increased robustness against photon-statistical and transport-induced effects compared to classical gate-based methods.

6.2 Evaluation on Imbalanced Test Dataset

The relatively modest FoM value reflects the fact that the neural network output represents a probabilistic score rather than a physically optimized scalar PSD parameter. Nevertheless, the ROC-AUC and recall metrics demonstrate strong discriminative performance, particularly for neutron detection, which is the operationally critical task in neutron–gamma separation.

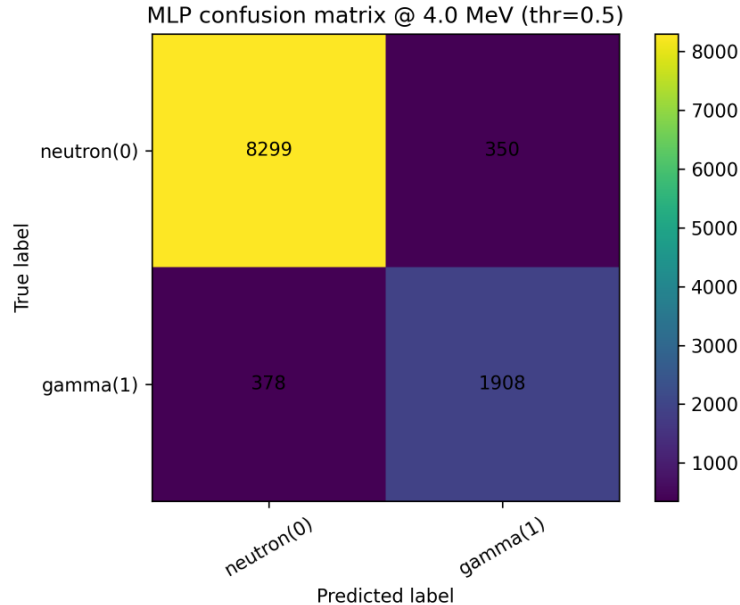


Figure 8: Confusion matrix for neutron–gamma classification using the MLP model.

The deterioration of charge-comparison (CC) performance at low deposited energies originates from the fundamental dependence of the method on late-time pulse integrals.

At low energies, the total number of detected scintillation photons is reduced, increasing the relative impact of photon-counting statistics and transport-induced timing fluctuations. Since the CC parameter compresses the entire waveform into a single tail-to-total ratio, these fluctuations directly broaden the CC distributions for both neutron and gamma events, leading to increased overlap and a reduced Figure of Merit. Figure 8 shows that the MLP achieves strong overall separation at 4.0 MeV with an accuracy of approximately 93%. Neutron identification is particularly robust: 8299 out of 8649 neutrons are correctly classified, corresponding to a neutron recall of 0.96%, indicating that missed-neutron events are rare under the fixed threshold $\tau = 0.5$. In contrast, gamma classification is slightly less efficient: 1908 out of 2286 gamma events are correctly identified (gamma recall 0.83%), with the dominant error mode being gamma events misclassified as neutrons (378 events). This asymmetric behavior is consistent with overlapping tail morphologies and event-to-event photon-statistical fluctuations that primarily affect gamma-like pulses. It also reflects the intentionally imbalanced test composition, in which robust neutron identification is a critical operational requirement.

7 Conclusion

In this thesis, neutron–gamma pulse shape discrimination in EJ-276 plastic scintillators has been investigated using a fully simulation-based and physics-driven framework. A dedicated GEANT4 simulation was developed to generate time-resolved scintillation pulses with explicit tracking of optical photon production and transport. This approach enables the construction of realistic waveform-level signals that naturally incorporate both intrinsic scintillation decay dynamics and transport-induced temporal dispersion, which are essential for meaningful pulse shape discrimination studies.

A detailed time-domain analysis confirms that neutron–gamma separation in plastic scintillators is primarily governed by differences in the delayed scintillation component. Neutron-induced pulses exhibit an enhanced late-time tail associated with dense ionization and slow molecular relaxation processes, whereas gamma-induced pulses are dominated by faster decay components. However, the discrimination-relevant tail region is also the most sensitive to photon-counting statistics and optical transport effects. Within the investigated energy range of 1–4 MeV, these effects introduce significant event-to-event fluctuations at low energies, leading to a degradation of classical pulse shape discrimination performance.

The classical charge comparison (CC) method was employed as a physically interpretable baseline. While CC performance improves with increasing deposited energy due to enhanced photon statistics, the results demonstrate that, at low energies, compressing the full waveform into a single scalar parameter leads to broadened neutron and gamma distributions and a reduced Figure of Merit. This energy-dependent sensitivity

highlights the intrinsic limitations of classical charge-based approaches under realistic noise and transport conditions.

To overcome these limitations, a waveform-level particle identification strategy based on a multilayer perceptron (MLP) was developed. By operating directly on full time-binned pulse shapes, the MLP exploits discrimination information distributed across both prompt and delayed components of the signal, rather than relying on a predefined integration window. The MLP demonstrates stable and reliable neutron–gamma separation across the entire studied energy range, maintaining high ROC–AUC values even in low-energy regimes where classical methods degrade. This stability indicates that the network learns physically meaningful pulse shape features rather than energy-specific artifacts.

Overall, this work demonstrates that time-domain, waveform-level analysis combined with machine learning provides a robust and physics-consistent alternative to classical pulse shape discrimination techniques in plastic scintillators. The results emphasize the importance of preserving the full temporal structure of scintillation signals and establish a solid foundation for extending PSD studies toward more realistic detector conditions and experimental validation. All simulation and analysis codes developed in this work are publicly available [9].

8 Future Perspectives

Future work will focus on extending the present framework toward more general and realistic conditions. Different neural network architectures, such as convolutional or recurrent models, may be compared to assess their efficiency in capturing time-domain scintillation features. Increasing the dataset size will reduce statistical uncertainties, particularly in the low-energy regime where photon-counting fluctuations dominate. Finally, incorporating a realistic electronics response and validating the simulation results against experimental data will be essential to assess the applicability of the proposed approach to real detector systems.

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